

EXPERIENCE

- **Citadel Asset Management** London, United Kingdom
Data Engineer → Engineering Manager 2022 - Present
 - **Market Data Infrastructure (Tick Pipeline):**
 - * Sole developer and owner of a department-wide tick data pipeline processing **30B+ market events per day**.
 - * Built a **low-latency, high-throughput** ingestion system in C++, Kafka, and Redis, leveraging concurrent and lock-free design patterns.
 - * Designed for resilience under peak market conditions: **zero-downtime rollout** and no production outages over 3+ years.
 - * Core market data feed powering research and trading workflows across the department.
 - **Research Data Platform (Storage & API Layer):**
 - * Sole architect and developer of the central research data platform, **adopted by 25 of 40 systematic trading desks** (~60%).
 - * Built on **PostgreSQL and Arrow Flight**, enabling high-performance bidirectional streaming APIs for large-scale data access.
 - * Implemented streaming reads/writes directly from database cursors, reducing memory overhead and improving throughput.
 - * Became the de facto standard for systematic research, **driving migration of all existing internal data APIs**.
 - **Leadership & Platform Transformation:**
 - * **Promoted from IC to Engineering Manager** within 3 years while continuing to own and ship production infrastructure.
 - * Leading **two teams (6 engineers)** across data onboarding and platform development.
 - * Reduced data onboarding time from **5+ months to under 30 days**, significantly accelerating dataset availability for research.
 - * Oversaw migration of **5,000+ ETL jobs**, standardising ingestion and reducing operational complexity.
 - * Offered the role of **Head of Data Engineering** for the department, expanding scope to a third team and full ownership of the data engineering function.
- **Bloomberg LP** London, United Kingdom
Senior Software Engineer 2019 - 2021
 - **Taxonomy and Metadata System:** Python, GraphQL, Redis, RabbitMQ, PostgreSQL, Data Pipeline
 - * **Architected ingestion pipelines** handling metadata for large-scale datasets across multiple asset classes.
 - * Added **support for sharding** to the ingestion pipeline, splitting work across 8 distributed shards and significantly increasing throughput.
 - * Team lead backup for a team of 7 engineers.

EDUCATION

- **University of St. Andrews** United Kingdom
MSc Statistics *Distinction, Deans' List, grade 17.9/20, September 2022*
 - **Modules:** Markov Chains, Bayesian Statistics, Multivariate Analysis, Statistical Modelling using GLM/GAMs, Quantitative risk analysis, Machine Learning
 - **Dissertation:** An Ensemble of HMM and KNN for signal generation in horse betting markets. Grade 19.3/20 — Highest grade in cohort
- **University of Sheffield** Remote, United Kingdom
Graduate Certificate in Statistics *Distinction, grade 90/100, July 2021*
 - **Modules:** Mathematical Methods for Statistics, Probability and Probability Distributions, Basic Statistics
- **Politechnica University of Timisoara** Timisoara, Romania
B.S.E in Computer Science *July 2014*
 - **Modules:** Software Engineering, Computer Architecture, Algorithms, Design Patterns, Programming Languages, Databases

SKILLS

- **Open Source:** [attrs-strict](#) — runtime validation library for Python attrs dataclasses, created and open-sourced at Bloomberg.
- **Architecture and technology:** Large-scale distributed systems, data pipelines, C++, Kafka, Redis, Arrow Flight, PostgreSQL, Python, gRPC, Airflow, lock-free concurrency
- **Mathematics:** Probability theory, Markov chains, Bayesian models, GLMs, statistical inference
- **Languages spoken:** English (IELTS 8.0), Romanian (native). Currently living in London.
- **Citizenships:** British, Romanian, Hungarian

PERSONAL PROJECTS

- **G-Research speed coding challenge:** (October 2021) Second place in Scotland at the university coding challenge organised by G-Research. Competed against teams of 4.
- **Cryptocurrency analysis** github.com/erikseulean/cryptoscraping: (January 2018) Scraped cryptocurrency data from multiple sources; implemented the Ichimoku indicator and correlation analysis between Google Trends and crypto prices. Python: Numpy, Pandas, Matplotlib, Multiprocessing, Jupyter.